Homework II, Theory of Computation 2026

Submission: **The deadline for Homework 2 is 23:59 on April 17.** Please submit your solutions on Moodle. Typing your solutions using a typesetting system such as LaTeX is strongly encouraged! If you must handwrite your solutions, write cleanly and with a pen. Messy and unreadable homeworks will not be graded. No late homeworks will be accepted.

Writing: Please be precise, concise and (reasonably) formal. Keep in mind that many of the problems ask you to provide a proof of a statement (as opposed to, say, just to provide an example). Therefore, make sure that your reasoning is correct and there are no holes in it. A solution that is hard/impossible to decipher/follow might not get full credit (even if it is in principle correct). You do not need to reprove anything that was shown in the class—just state clearly what was proved and where.

Collaboration: These problem sets are meant to be worked on in groups of 2–4 students. Please submit only one writeup per team—it should contain the names of all the students. You are strongly encouraged to solve these problems by yourself. If you must, you may use books or online resources to help solve homework problems, but you must credit all such sources in your writeup and you must never copy material verbatim. Even though only one writeup is submitted, it is expected that each one of the team members is able to fully explain the solutions if requested to do so.

Grading: Each of the two problems will be graded on a scale from 0 to 5.

Warning: Your attention is drawn to the EPFL policy on academic dishonesty. In particular, you should be aware that copying solutions, in whole or in part, from other students in the class or any other source (e.g., ChatGPT) without acknowledgment constitutes cheating. Any student found to be cheating risks automatically failing the class and being referred to the appropriate office.

Homework 2

1 Classify each of the following languages into *one* of the following three categories: decidable, undecidable but recognizable, unrecognizable. Prove your answers.

(a) $L_1 = \left\{ \langle M \rangle \;\middle|\; \begin{array}{l} M \text{ is a TM} \\ \exists\, w_1 \neq w_2 \text{ with } |w_1| = |w_2| \\ \text{such that } M \text{ accepts } w_1 \text{ and rejects } w_2 \end{array} \right\}$

(b) $L_2 = \left\{ \langle M \rangle \;\middle|\; \begin{array}{l} M \text{ is a TM} \\ \forall\, n \geq 0\; \exists\, w \in \Sigma^n \\ \text{such that } M \text{ halts on } w \end{array} \right\}$

2 A property of Turing-recognizable languages $\mathcal{P}$ is a set whose elements are Turing-recognizable languages over a fixed alphabet Σ. The property $\mathcal{P}$ is *semantic* if it depends only on the language recognized, not on the particular machine recognizing it. Equivalently, for all TMs M, N:

$$L(M) = L(N) \implies \big(L(M) \in \mathcal{P} \iff L(N) \in \mathcal{P}\big) \ .$$

We say that $\mathcal{P}$ is *non-trivial* if there exist TMs M, N such that $L(M) \in \mathcal{P}$ and $L(N) \notin \mathcal{P}$. Examples of non-trivial semantic properties of Turing-recognizable languages include "Does M terminate on a given input $x \in \Sigma^n$?" (halting problem), "Does M terminate on all inputs $x \in \Sigma^n$?", or "Is $L(M)$ empty?". The objective of this problem is to obtain a proof of the following result, known as **Rice's theorem**:

Theorem. *Let $\mathcal{P}$ be a non-trivial semantic property of Turing-recognizable languages, and define $L(\mathcal{P}) := \{\langle M \rangle \mid L(M) \in \mathcal{P}\}$. Then, $L(\mathcal{P})$ is undecidable.*

(a) Give an example of a property of Turing-recognizable languages that is not semantic.

(b) Let $\mathcal{P}$ be a non-trivial semantic property of Turing-recognizable languages. Show that there exists a TM Y such that **exactly one** of $\emptyset$ and $L(Y)$ is in $\mathcal{P}$.

(c) Given $\langle M, w \rangle$ and $\langle Y \rangle$, construct a TM $N_{M,w}$ such that:

- if M accepts w, then $L(N_{M,w}) = L(Y)$.
- if M does not accept w, then $L(N_{M,w}) = \emptyset$.

(d) Conclude that $L(\mathcal{P})$ is undecidable.